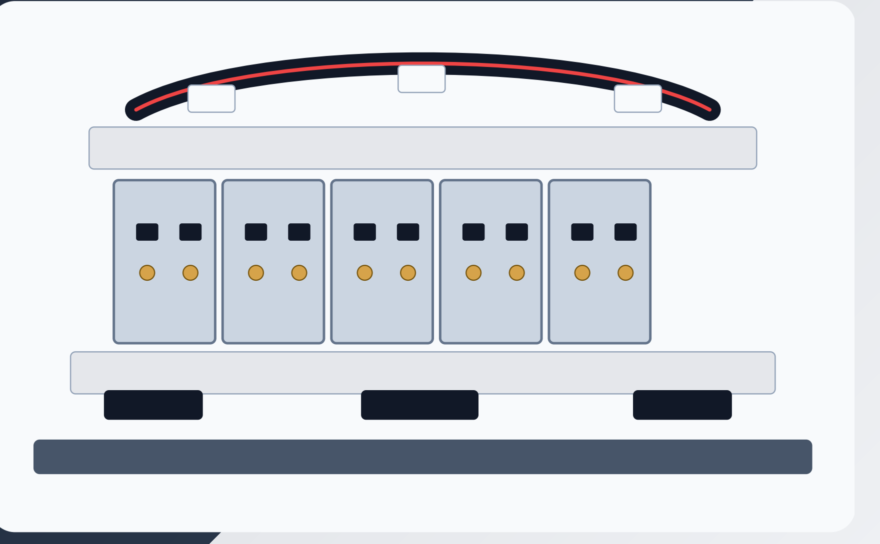
Static Robo HMI Test Jig Build Guide

Self-contained PDF with visual build instructions

Use this document to start designing and building a modular HMI electrical test fixture with aluminum extrusion, removable connector panels, routed harnesses, safety protection, and a repeatable test flow.

Frame Panels Harness Control Box Testing

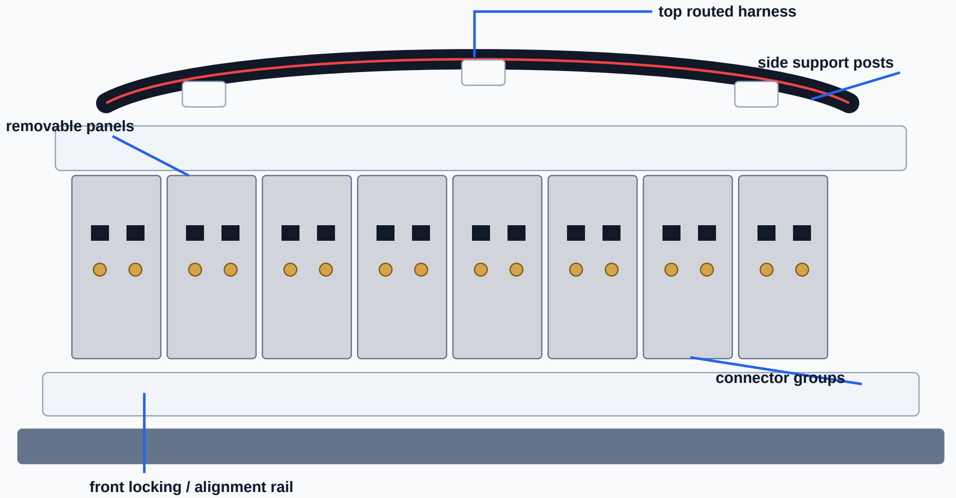


Static-Robo-HMI-test-JIG

Generated build guide

1. Reference Fixture — What to Copy

Annotated visual reconstruction



The live GitHub attachment URLs could not be resolved inside the build sandbox, so this PDF uses local, self-contained illustrations based on the visible reference fixture.

Main design intent: build a rigid modular fixture that repeatedly connects to an HMI or harness assembly, runs electrical checks, and gives the operator a clear pass/fail result.

Visible features to reproduce

- Aluminum extrusion base and top frame.
- Multiple removable vertical connector panels.
- Repeated black connector blocks and round metal test connectors.
- Harness routed across the top with fixed clamps.
- Left/right support posts and hard stops.
- Front alignment/locking hardware for repeatable loading.
- Serviceable modular construction with exposed fasteners.

Practical output

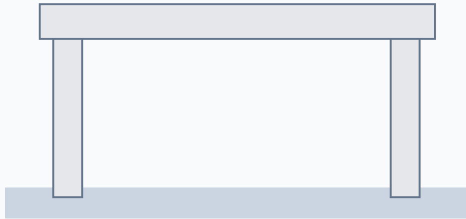
- One panel per connector group.
- One controller channel map per panel.
- One terminal-block row per harness branch.
- Clear labels on front and rear of every connector.

2. Step-by-Step Build Storyboard

1 Base frame

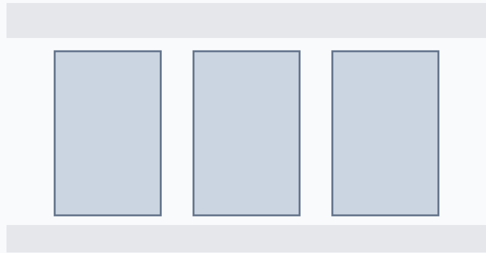
- Cut T-slot extrusion to length.
- Square the frame before tightening.
- Add feet or bench mounting plates.

2 Vertical posts



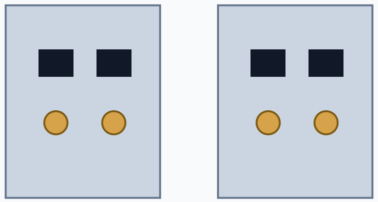
- Install left/right supports.
- Add top rail and rear cable rail.
- Check frame stiffness.

3 Panel plates



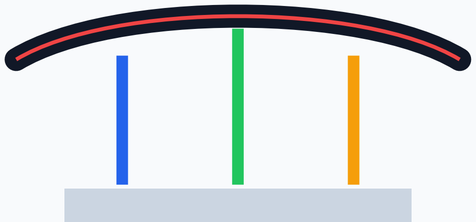
- Machine removable plates.
- Add captive screws or dowels.
- Keep panel widths repeatable.

4 Connectors



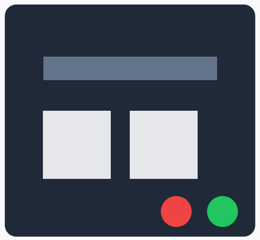
- Install panel-mount connectors.
- Use keyed, rated connectors.
- Label before wiring.

5 Harness



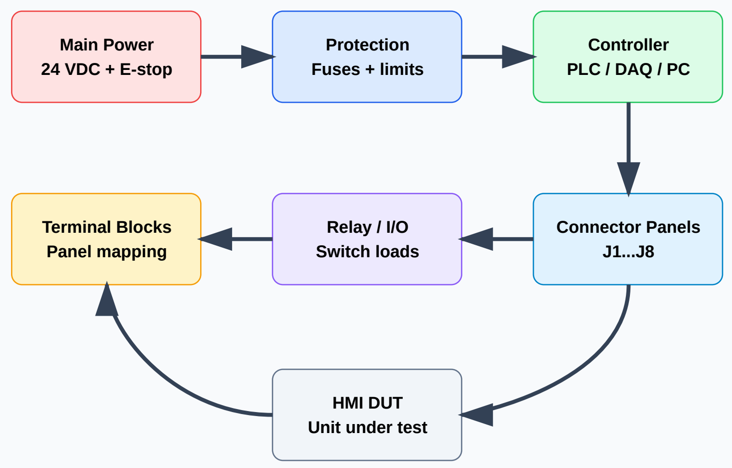
- Route across top/rear rail.
- Clamp every branch point.
- Leave service loops.

6 Control box



- Add power, E-stop, PLC/DAQ.
- Terminate to labeled blocks.
- Separate AC and DC wiring.

3. Electrical Architecture



Electrical design checklist

- Create a full I/O list before wiring.
- Map every connector pin to a terminal block and controller channel.
- Fuse each power branch.
- Use relay isolation when switching external loads.
- Add E-stop and verified safe power removal.
- Separate mains, 24 VDC, analog, and communication wiring.
- Use shielded cable for analog or communication lines.

Test	Method	Pass limit
Continuity	Measure pin-to-pin resistance	< 1 ohm typical
Short check	Check adjacent pins and rails	Open circuit
Power rail	Measure energized rail	23-25 VDC for 24 V rail
Current	Measure load draw	Below product limit
Functional I/O	Toggle input/output	Expected state change

4. Mechanical and Wiring Details

Frame hardware

- Use 30 mm or 40 mm T-slot extrusion.
- Use corner brackets and T-nuts.
- Add side posts to stop flexing.
- Use dowel pins for fixture repeatability.
- Add front stops and clamp blocks.

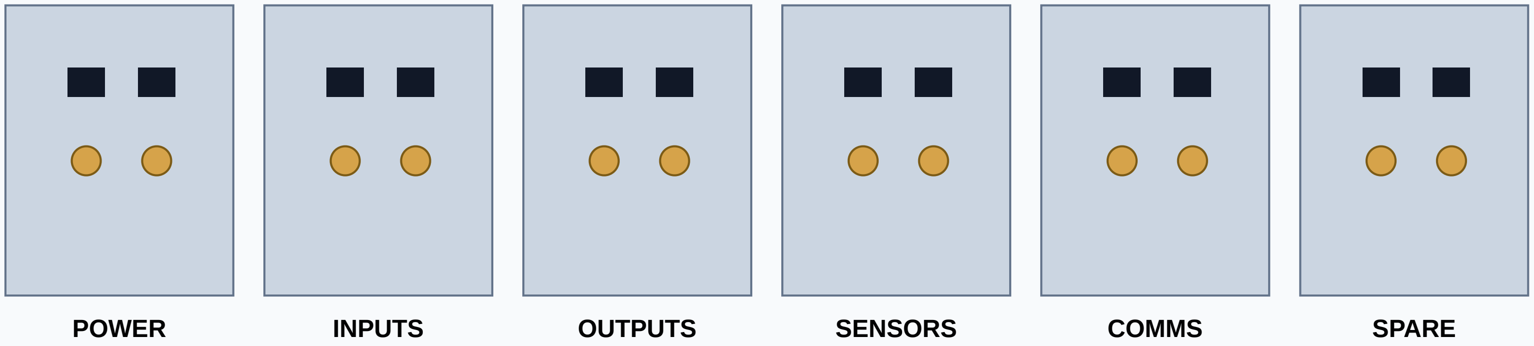
Panel hardware

- Use machined aluminum for final panels.
- Use acrylic/Delrin for prototype panels.
- Make panels removable.
- Group connectors by function.
- Put labels on both sides.

Harness hardware

- Use cable duct or clamp blocks.
- Bundle power, I/O, and communication separately.
- Use ferrules at terminal blocks.
- Add strain relief near every connector.
- Keep service loops for repair.

Connector panel layout example

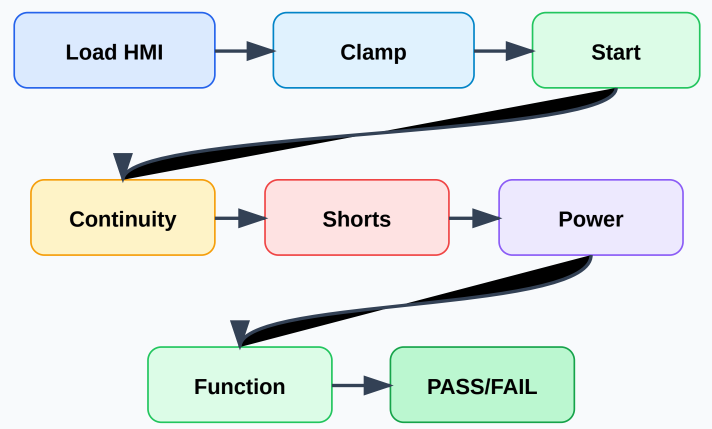


5. Build Procedure

Assembly sequence

1. Freeze connector pinout and I/O list.
2. Cut and square the aluminum extrusion base.
3. Install vertical support posts and top harness rail.
4. Machine removable connector panels.
5. Install panel-mount connectors and test sockets.
6. Add front locating blocks, hard stops, and clamps.
7. Build the control box with power, E-stop, fuses, PLC/DAQ, relays, and terminal blocks.
8. Route harness bundles across the top/rear rail.
9. Terminate every wire and label both ends.
10. Run dry continuity checks before powering.
11. Power one section at a time.
12. Run full functional test and record results.

Operator flow



Safety: review the final electrical design with a qualified electrical engineer or technician. Verify E-stop behavior, grounding, fusing, current limits, and operator-safe terminals before production use.

6. Starter BOM and Pinout Templates

Starter bill of materials

Category	Items
Frame	T-slot extrusion, brackets, T-nuts, base plates, feet
Panels	Aluminum/Delrin plates, captive screws, dowel pins
Connectors	Panel-mount connectors, mating plugs, pogo pins, sockets
Wiring	Hookup wire, shielded cable, ferrules, labels, strain reliefs
Control	PLC/DAQ/microcontroller, relay modules, terminal blocks
Power/Safety	24 VDC PSU, main switch, E-stop, fuses, grounding bar
Operator UI	Start/reset buttons, pass/fail lamps, buzzer, display

Pinout template

Connector	Pin	Signal	Rating	Test	Pass limit
J1	1	+24 VDC	24 V	Voltage	23-25 V
J1	2	0 VDC	Return	Continuity	< 1 ohm
J2	1	Button 1	24 V input	Toggle	State changes
J3	1	LED 1	24 V output	Command	Output ON

Recommended first prototype

Build one panel

Use one connector group and one small removable plate.

Wire one I/O set

Connect power, one input, one output, and one test point.

Run one pass/fail test

Validate continuity, shorts, power, and one function.